# Lab 4

By

Caden Burnett

10/23/2025

**“On my honor as an Aggie, I have neither given nor received unauthorized aid on this academic work.”**

This is entirely my own work or—if authorized by my instructor—the work of the team I was assigned to--except as documented in a bibliography or acknowledged below.

Signature /Caden Burnett/

*You may sign and scan the cover page, add a signature after converting to PDF, or type your name like this: */Joe Aggie/*

Artificial Intelligence (AI) Acknowledgement: Chat GPT helped me write

Submitted to fulfill the requirement for Lab X
ISEN 350, Fall 2025
Department of Industrial & Systems Engineering
Texas A&M University

# 1. Table of Contents

Table of Figures ..... 1
Table of Tables ..... 2
Introduction ..... 3
Methods ..... 3
Methods for Exercise 1 ..... 3
Methods for Exercise 2 ..... 4
Methods for Exercise 3 ..... 5
Results ..... 5
Results for Exercise 1 ..... 5
Results for Exercise 2 ..... 10
Results for Exercise 3 ..... 13
Results for Sample Size of 5 for Each Subgroup ..... 13
Results for Sample Size of 15 for Each Subgroup ..... 15
Results for Sample Size of 30 for Each Subgroup ..... 18
Conclusions ..... 21

# Table of Figures

Figure 1. Exercise 1 : Run Chart ..... 6
Figure 2. Exercise 2 : Histogram ..... 7
Figure 3. Exercise 1 : Box Plot ..... 8
Figure 4. Exercise 2 : Probability Plot ..... 9
Figure 5. Exercise 2 : Histogram ..... 11
Figure 6. Exercise 2 : Probability Plot ..... 12
Figure 7. Exercise 3 : Histogram of Stacked Raw Data for n=5 ..... 13
Figure 8. Exercise 3 : Histogram of Sample Size Averages for n=5 ..... 14
Figure 9. Exercise 3 : Probability Plot Comparison for n=5 ..... 15
Figure 10. Exercise 3 : Histogram of Stacked Data for n=15 ..... 16
Figure 11. Exercise 3 : Histogram of Sample Averages for n=15 ..... 17
Figure 12. Exercise 3 : Probability Plot Comparison for n=15 ..... 18
Figure 13. Exercise 3 : Histogram of Stacked Data for n=30 ..... 19

Figure 14. Exercise 3 : Histogram of Sample Averages for n=30 ..... 20
Figure 15. Exercise 3 : Probability Plot Comparison for n=30 ..... 21

## Table of Tables

Table 1. Exercise 1: Descriptive Statistics ..... 6
Table 2. Exercise 2 : Descriptive Statistics ..... 10

# 1. Introduction

This lab investigated several fundamental concepts in probability and statistics as they relate to engineering processes. The first exercise focused on descriptive statistics and graphical tools, including histograms, boxplots, probability plots, and run charts, in order to assess process stability and evaluate whether data followed a normal distribution. The second exercise examined the theoretical and experimental distribution of a linkage length formed by the sum of independent normally distributed shaft lengths, comparing the theoretical parameters with simulated results and confidence intervals. The third exercise demonstrated the Central Limit Theorem using data generated from a skewed Gamma distribution, showing how the distribution of sample means approaches normality as the subgroup size increases. Together, these exercises provided practical applications of statistical methods for evaluating process performance, distributional assumptions, and the behavior of averages.

# 2. Methods

## 2.1 Methods for Exercise 1

Bottle weight data were compiled into a single column in Minitab and analyzed using descriptive statistics to obtain measures of central tendency, spread, and distribution shape. A histogram and boxplot were generated to evaluate the overall distribution and to identify possible outliers. A probability plot was constructed to assess normality. A run chart was produced to examine variation across time and to detect clustering, mixtures, or oscillations.

If the sample data were shown to be approximately normal, Eq.1 is used to find the probability of a glass weighing outside of the specification limits.

$$Probability(defect) = 1 - \int_{LSL}^{XSL} N(\bar{X}, \bar{\sigma}) d\bar{X} \tag{1}$$

$\bar{X}$ is the sample mean, $\bar{\sigma}$ is the sample standard deviation, and USL and LSL are the specification limits (320g, 330g). This computation will provide an estimate for process defects in parts per million shown in Eq.2.

$$Defects(ppm) = 1{,}000{,}000 \times Probability(defect) \tag{2}$$

The result of this calculation will be compared to the Six Sigma Quality standard of 3.4 parts per million. Additionally, the sigma level for the process will be found using equation 3.

$$Sigma\ level = \Phi^{-1}(1 - \alpha) \tag{3}$$

This equation finds the inverse normal of $1 - \alpha$ where $\alpha = \frac{Probability(defect)}{2}$.

The resulting metrics will be used to evaluate the process quality with respect to the six sigma quality levels.

## 2.2 Methods for Exercise 2

The theoretical distribution of the linkage length $L$is normal because the sum of independent normal random variables is also normally distributed. The mean of this distribution is obtained by adding the means of the three shafts. The variance of $L$is calculated as the sum of the variances of the individual shaft distributions, and the standard deviation is the square root of this total variance.

To determine the distribution of the linkage length experimentally, first, 500 random measurements will be generated for each distribution. For each index, $x_1 + x_2 + x_3$ will give an experimental L. The set of 500 experimental L values will be graphically and statistically analyzed and compared to the theoretical distribution

## 2.3 Methods for Exercise 3

Random data were generated from a Gamma distribution with a shape parameter of 3.0, a scale parameter of 1.0, and a threshold of 0.0 using Minitab. Five columns of 100 values each were created and then stacked into a single column to represent the overall distribution of the original data. A histogram of these stacked values was produced to visualize the shape of the parent distribution.

Next, sample means were calculated by grouping the generated values into subgroups of size five. The row statistics function in Minitab was used to compute the mean of each subgroup, resulting in 100 sample means. These sample means were stored in a new column, and a histogram was generated to illustrate the distribution of the averages.

Finally, the distribution of the sample mean was assessed using the Individual Distribution Identification tool. The sample mean data were fit to several candidate distributions, including the normal, gamma, Weibull, and exponential, and the resulting probability plots and p-values were examined. This step provided evidence regarding whether the distribution of the sample mean approximates a normal distribution, as predicted by the Central Limit Theorem for a sample size of five.

The same steps were repeated for additional subgroup sizes in order to examine the effect of sample size on the distribution of the sample mean, consistent with the Central Limit Theorem.

# 3. Results

## 3.1 Results for Exercise 1

The statistical analysis of the glass bottle sample data was used to determine the statistics in Table 1.

*Table 1. Exercise 1: Descriptive Statistics*

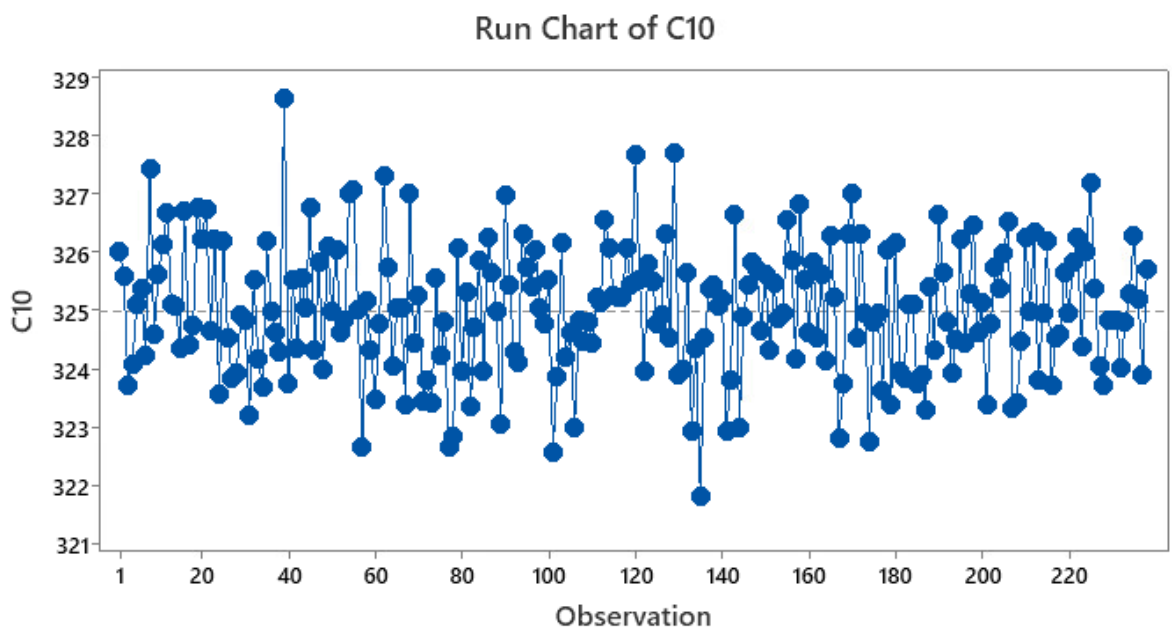
## Statistics

| Variable | N | N* | Mean | SE Mean | StDev | Variance | Minimum | Q1 | Median | Q3 |
|---|---|---|---|---|---|---|---|---|---|---|
| C10 | 238 | 0 | 325.006 | 0.0726729 | 1.12114 | 1.25696 | 321.808 | 324.224 | 324.976 | 325.756 |

| Variable | Maximum | IQR | Mode | N for Mode | Skewness | Kurtosis |
|---|---|---|---|---|---|---|
| C10 | 328.624 | 1.53134 | * | 0 | 0.07 | -0.05 |

The sample mean of 325.006 and the sample standard deviation of 1.12114 will serve as the parameters of the normal distribution if the data is in fact normal.

To determine if the process is stable and under statistical control, a run chart of bottle weights was constructed in Figure 1.




| | | | |
|---|---|---|---|
| Number of runs about median: | 125 | Number of runs up or down: | 166 |
| Expected number of runs: | 120.0 | Expected number of runs: | 158.3 |
| Longest run about median: | 11 | Longest run up or down: | 3 |
| Approx P-Value for Clustering: | 0.742 | Approx P-Value for Trends: | 0.882 |
| Approx P-Value for Mixtures: | 0.258 | Approx P-Value for Oscillation: | 0.118 |

*Figure 1. Exercise 1 : Run Chart*

Graphical analysis of the run chart shows bottle weights fluctuating randomly around the median without evidence of clustering, mixtures, or trends. The p-values for clustering (0.742), mixtures (0.258), and trends (0.882) all exceed the 0.05 threshold, indicating no statistically significant departures from randomness. Although the p-value for oscillation

(0.118) is lower than the others, it remains above the significance threshold and does not provide strong evidence of cyclical behavior. These results suggest that the observed variation is consistent with common cause variation rather than assignable causes.

A histogram of bottle weights was constructed to provide an initial assessment of distributional shape in Figure 2.

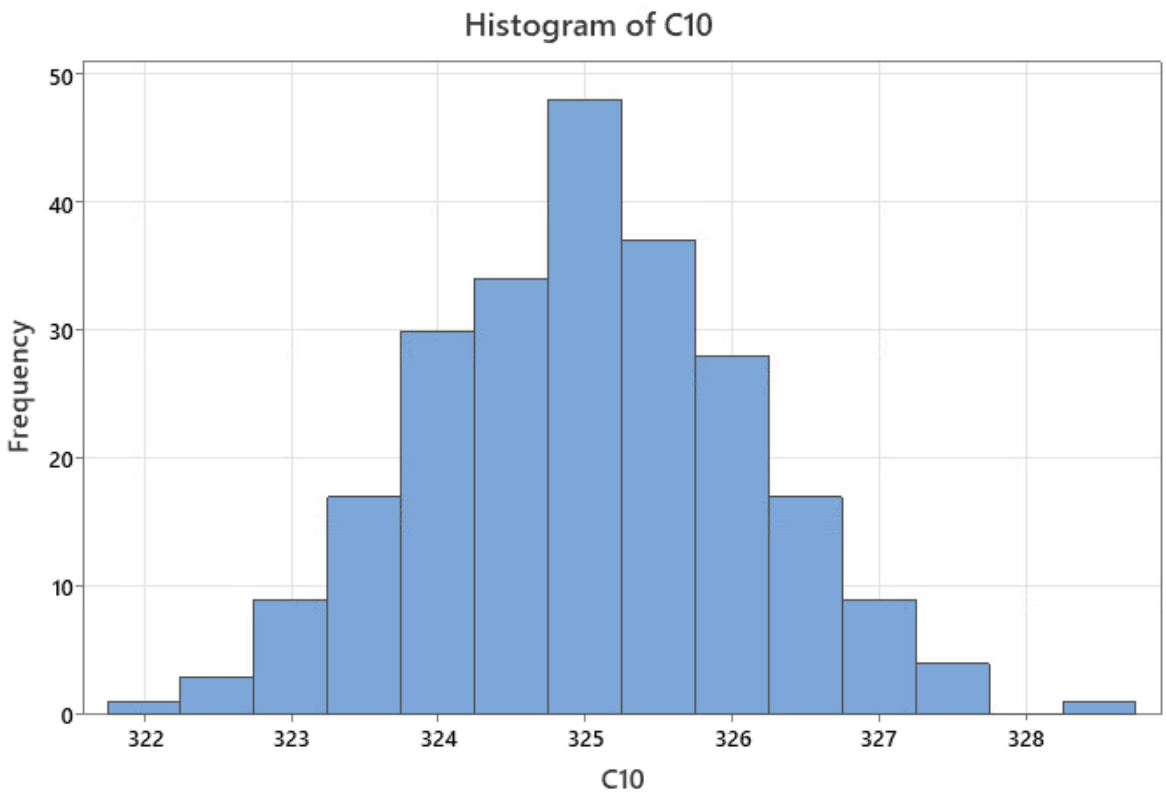



*Figure 2. Exercise 2 : Histogram*

The histogram appears symmetric and bell-shaped, centered close to 325 g, with no evidence of skewness or bimodality. This suggests that the production process is well-centered on the target mean and that variation is consistent with a normal distribution.

A boxplot was prepared to evaluate spread and to identify potential outliers in Figure 3.

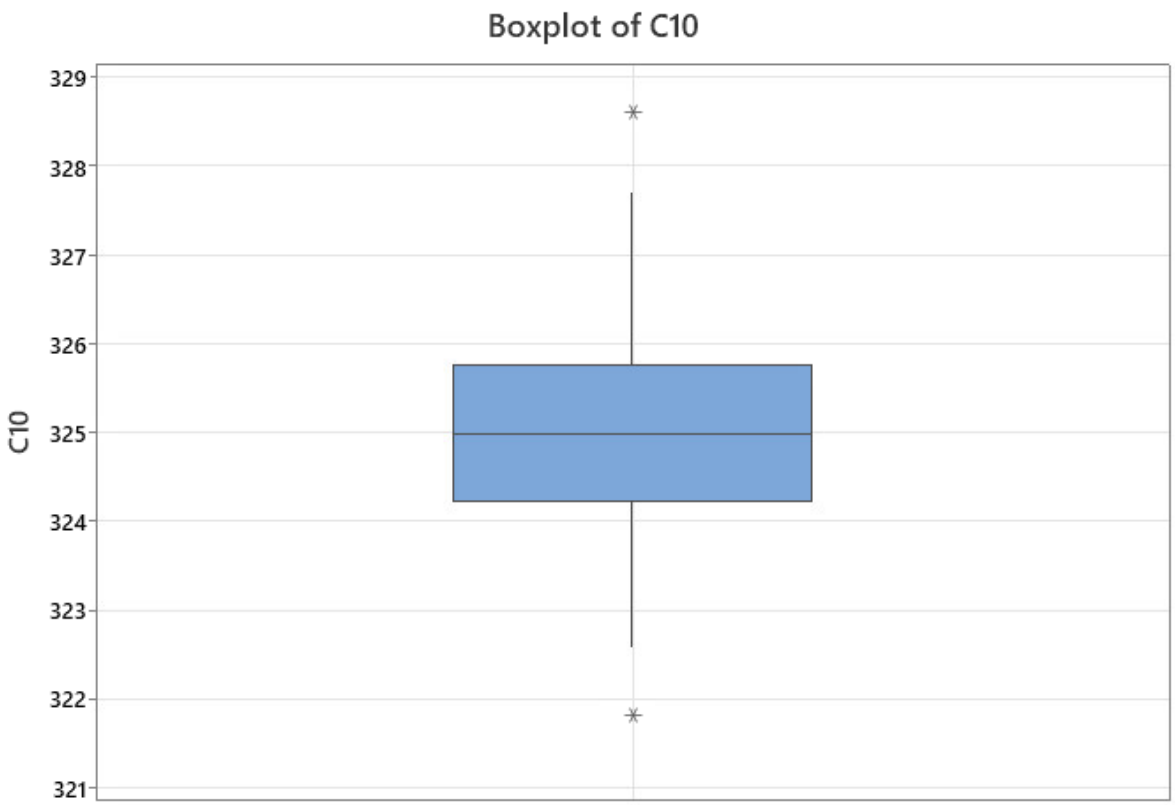



*Figure 3. Exercise 1 : Box Plot*

The boxplot shows a median at approximately 325 g with interquartile limits spanning a narrow range. Two outliers are visible, one on the lower side near 321 g and one on the upper side near 329 g. Because these values remain within specification limits, they are not indicative of a process shift and are more consistent with random variation.

A probability plot was generated to formally test normality in Figure 4.

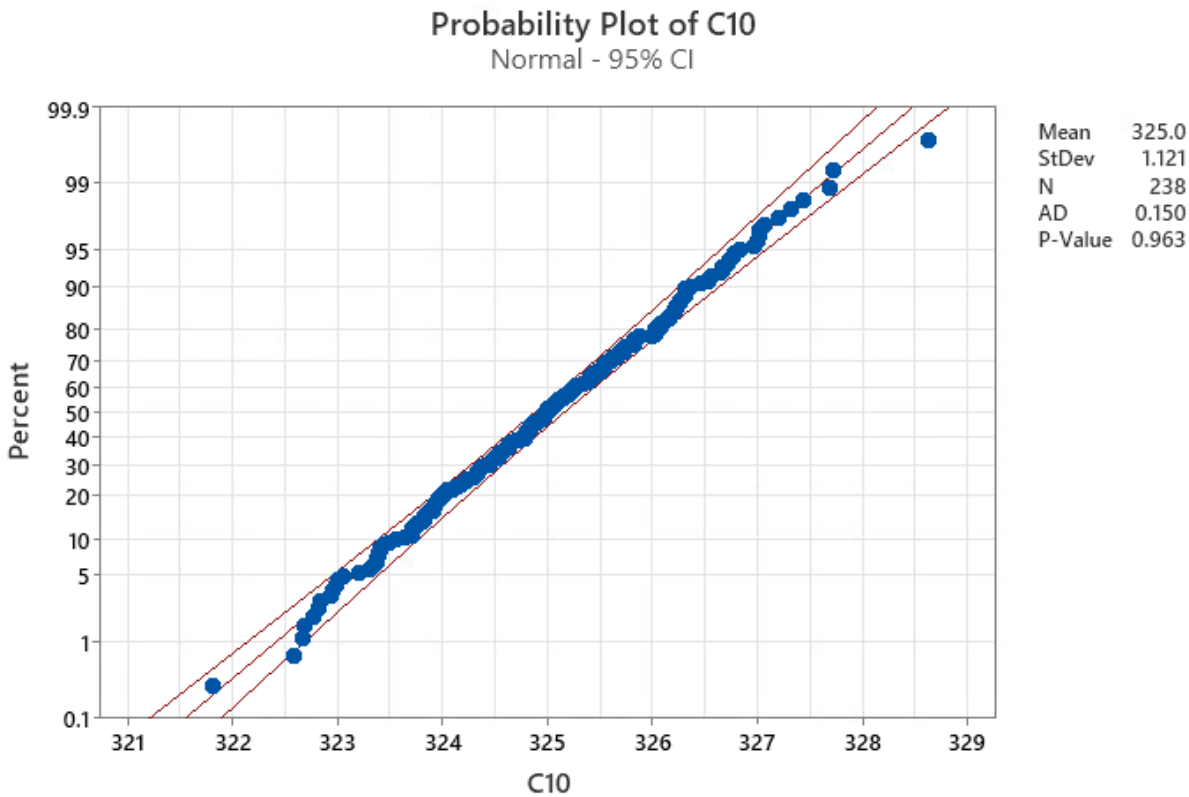



*Figure 4. Exercise 2 : Probability Plot*

The data points fall closely along the reference line, the Anderson–Darling statistic is low (0.150), and the p-value is high (0.963). These results indicate that the assumption of normality cannot be rejected, confirming that the data follow a normal distribution. This supports the use of the sample mean and standard deviation as parameters for the normal model when estimating the probability of bottles meeting specifications.

The calculation for probability of a glass bottle being defective is evaluated using Eq.1 with in terms of the sample mean and standard deviation parameters.

$$P(defect) = 1 - \int_{320}^{330} N(325.006, 1.12114)\, d\,\bar{X} = 8 \times 10^{-6} \qquad (1)$$

Multiplying this result by 1,000,000 gives around 8.2 defects in parts per million. The resulting sigma level is around $4.47\sigma$. Accounting for the $+1.5$ long term shift in process variability, the process sigma level is around $5.97\sigma$.

## 3.2 Results for Exercise 2

The theoretical distribution of the linkage length is L~N(0.545,0.738).

Analysis of the set of experimental L measurements concludes the statistical data in Table 2.

*Table 2. Exercise 2 : Descriptive Statistics*

**Statistics**

| Variable | N | N* | Mean | SE Mean | StDev | Variance | Minimum | Q1 | Median | Q3 |
|---|---|---|---|---|---|---|---|---|---|---|
| Sum | 500 | 0 | 79.0087 | 0.0328374 | 0.734266 | 0.539146 | 77.0585 | 78.4994 | 78.9730 | 79.5651 |

| Variable | Maximum | IQR | Mode | N for Mode | Skewness | Kurtosis |
|---|---|---|---|---|---|---|
| Sum | 81.2097 | 1.06571 | * | 0 | -0.03 | -0.28 |

The mean of 79.0087 and the standard deviation of 0.734266 will be the parameters for the experimental normal distribution. This data produces a 95% confidence interval of (78.946,79.074).

A histogram of the estimated linkage lengths was constructed to provide an initial visual assessment of the distribution in Figure 5.

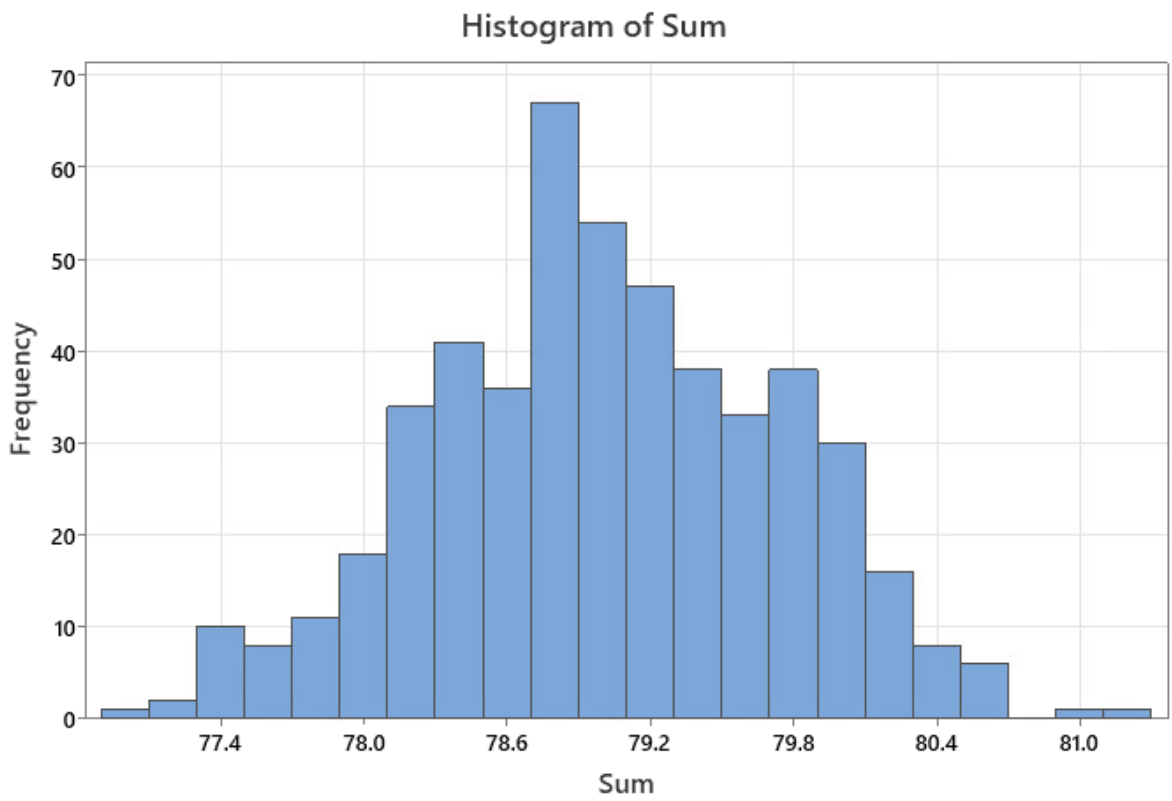



*Figure 5. Exercise 2 : Histogram*

The histogram appears symmetric and bell-shaped, centered close to 79 cm, with no evidence of skewness or multimodality. This pattern suggests the data are consistent with a normal distribution, as predicted theoretically.

A probability plot was then generated to test the assumption of normality more formally in Figure 6.

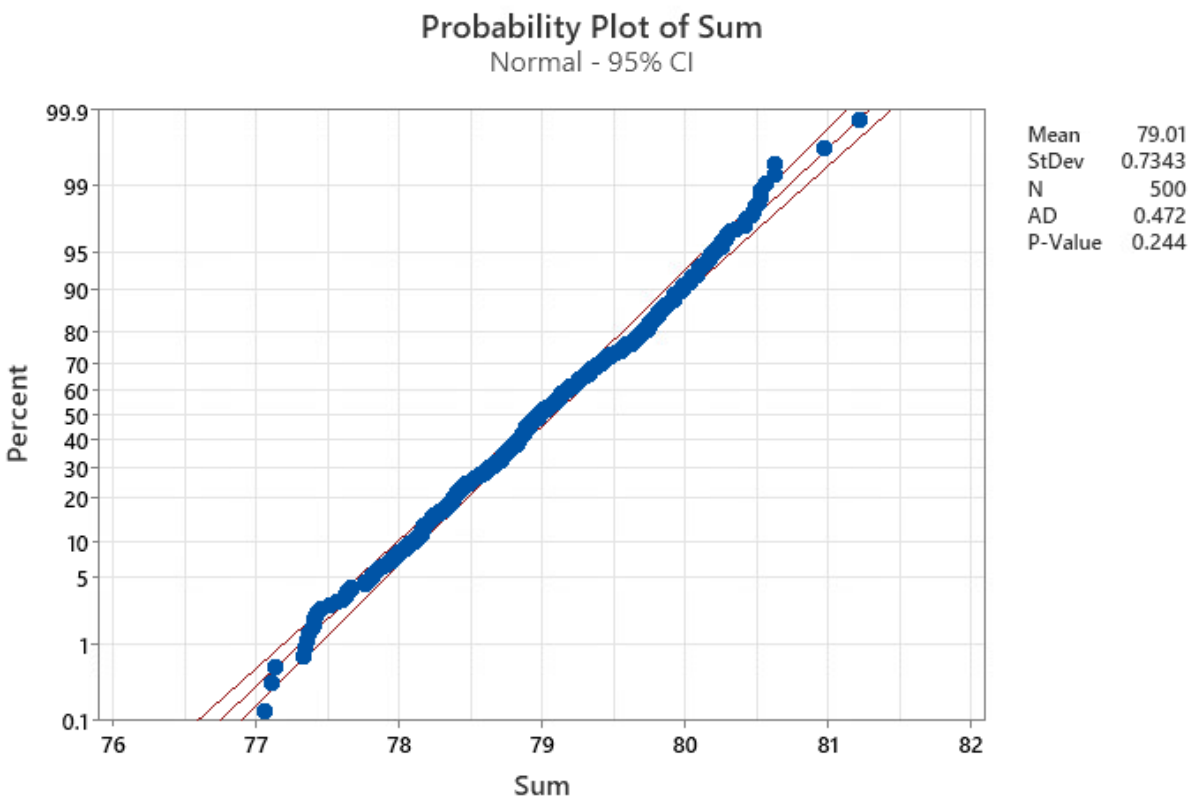



*Figure 6. Exercise 2 : Probability Plot*

The points fall close to the reference line and remain within the confidence bands, with an Anderson–Darling statistic of 0.472 and a p-value of 0.244. Since the p-value is well above 0.05, there is no evidence to reject normality, indicating that the simulated data provide a good fit to the normal model.

The descriptive statistics of the simulation give a sample mean of 79.01 cm and a sample standard deviation of 0.734 cm. These values align closely with the theoretical parameters of $\mu = 79$ cm and $\sigma = 0.738$ cm, confirming that the experimental results closely match the expected theoretical distribution. This is supported by the theoretical mean of 79 lies inside the 95% confidence interval.

## 3.3 Results for Exercise 3

### 3.3.1 Results for Sample Size of 5 for Each Subgroup

The histogram of the stacked raw data in Figure 7 shows the distribution of the Gamma-generated values for the 5 combined sets of sample data.

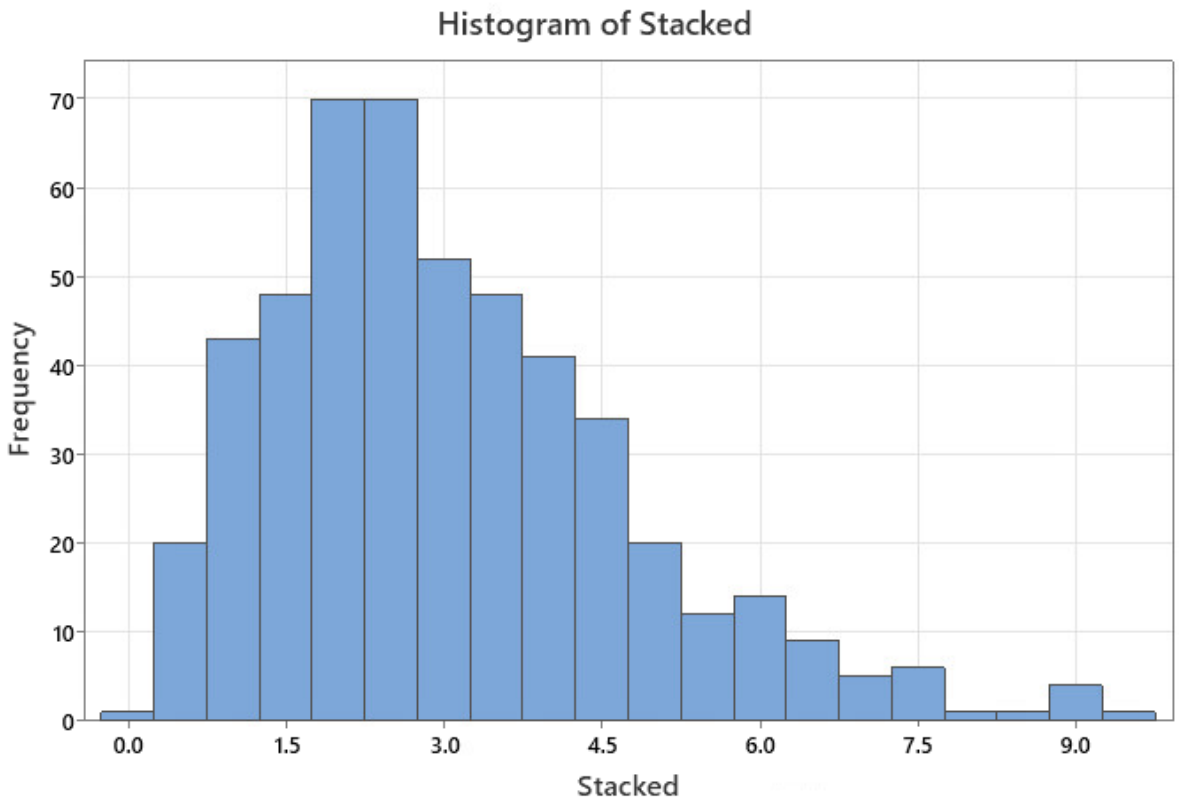



*Figure 7. Exercise 3 : Histogram of Stacked Raw Data for n=5*

The shape is positively skewed with a long right tail, which confirms that the parent population does not follow a normal distribution. This provides the baseline for comparison with the distribution of the sample means.

The histogram of the subgroup means in Figure 8 displays the averages of samples of size five.

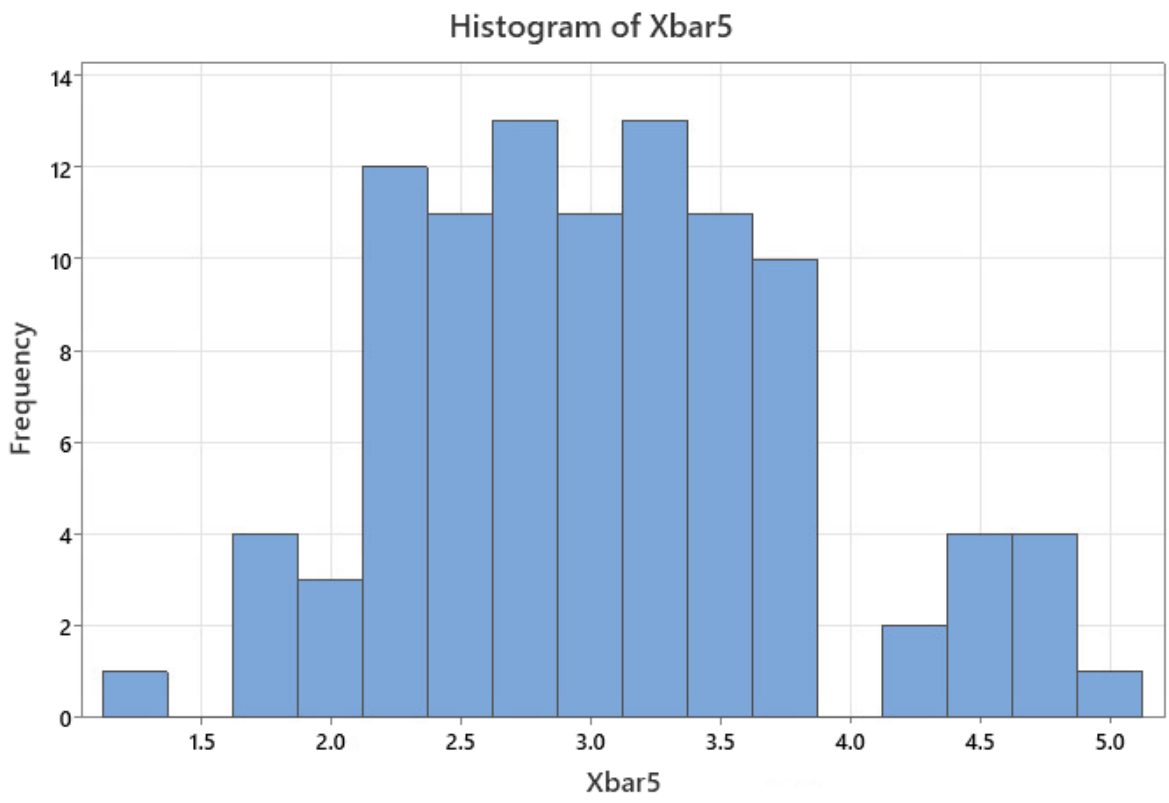



*Figure 8. Exercise 3 : Histogram of Sample Size Averages for n=5*

Unlike the raw data, the histogram is more symmetric and resembles a bell-shaped curve, with most values concentrated around three. This reduction in skewness supports the Central Limit Theorem prediction that the distribution of sample means tends toward normality, even when the parent distribution is skewed.

The probability plots of the sample means in Figure 9 formally test distributional fit.

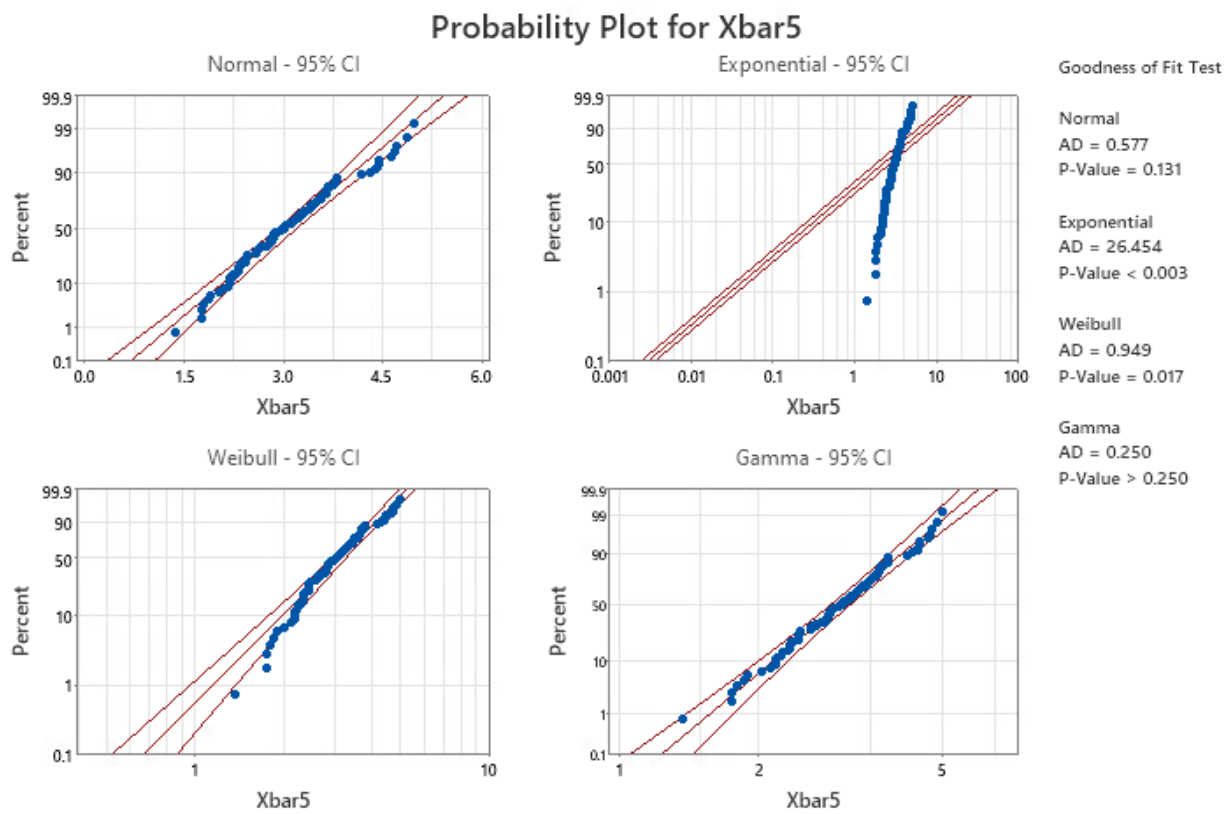



*Figure 9. Exercise 3 : Probability Plot Comparison for n=5*

The normal probability plot shows points lying close to the reference line with a p-value of 0.131, indicating that normality cannot be rejected. The gamma fit also provides acceptable alignment, with a p-value greater than 0.25. By contrast, the exponential and Weibull plots yield low p-values, demonstrating poor fit. These results show that the sample mean distribution is adequately described by the normal distribution and fits better than the exponential or Weibull alternatives.

### 3.3.2 Results for Sample Size of 15 for Each Subgroup

The histogram of the stacked raw data in Figure 10 shows the skewed shape of the parent Gamma distribution for 10 combined sets of sample data.

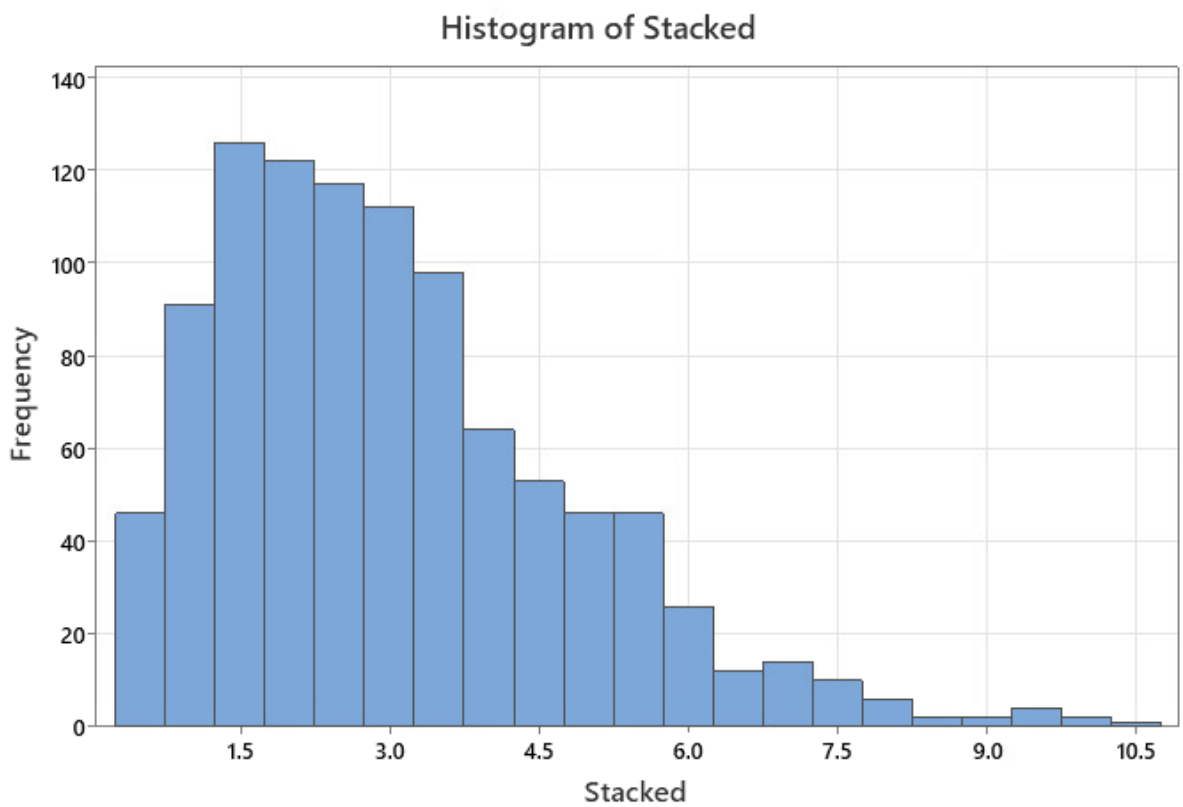



*Figure 10. Exercise 3 : Histogram of Stacked Data for n=15*

The data are concentrated between 1 and 4 with a long right tail, confirming that the underlying population is not normal. This provides a baseline for comparison with the sample mean distribution at a larger sample size.

The histogram of the sample means for subgroups of size 10 in Figure 11 shows a distribution that is more symmetric than the parent data.

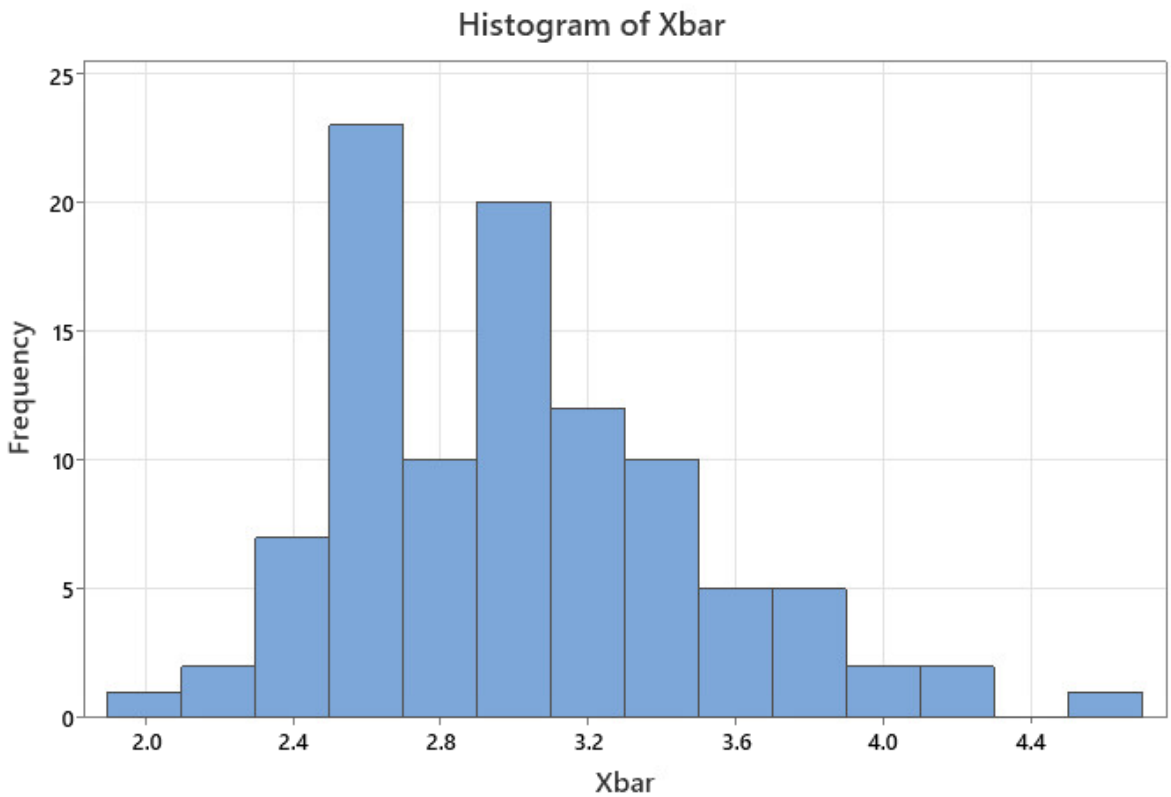



*Figure 11. Exercise 3 : Histogram of Sample Averages for n=15*

The majority of subgroup averages fall between 2.5 and 3.5, with fewer values in the tails. Compared to the $n = 5$ case, the histogram demonstrates reduced variability and appears closer to a bell-shaped curve, reflecting the effect of increasing sample size on the distribution of the mean.

The probability plots for the sample means in Figure 12 further illustrate this trend.

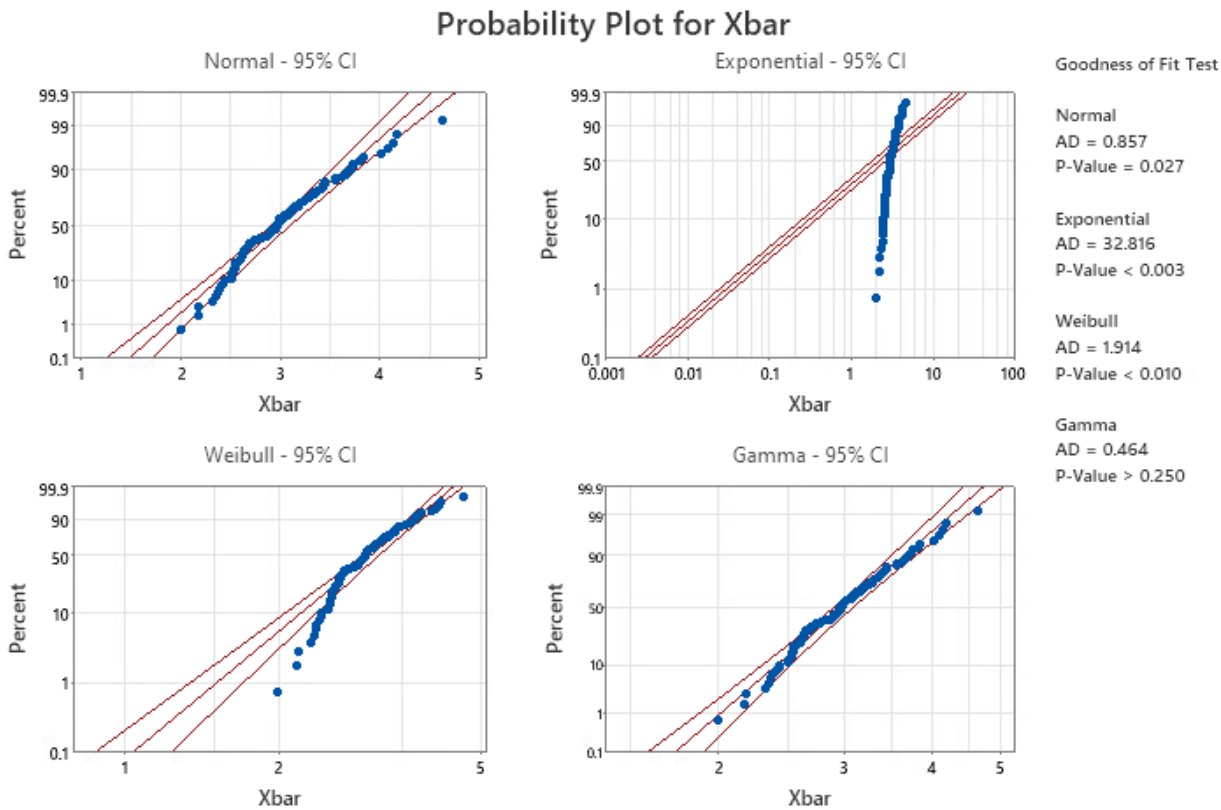



*Figure 12. Exercise 3 : Probability Plot Comparison for n=15*

The normal probability plot shows a relatively straight alignment, although the p-value of 0.027 indicates only marginal support for normality. The gamma distribution still provides a very good fit, with a p-value greater than 0.25. By contrast, the exponential and Weibull fits yield very low p-values, showing poor alignment. Compared to the $n = 5$ results, the $n = 10$ case shows improved symmetry and closer agreement with the normal distribution, but the gamma distribution remains a slightly better fit.

### 3.3.3 Results for Sample Size of 30 for Each Subgroup

The histogram of the parent Gamma distribution in Figure 13 again highlights the skewed nature of the raw data, with most values between 1.5 and 3.5 and a long right tail.

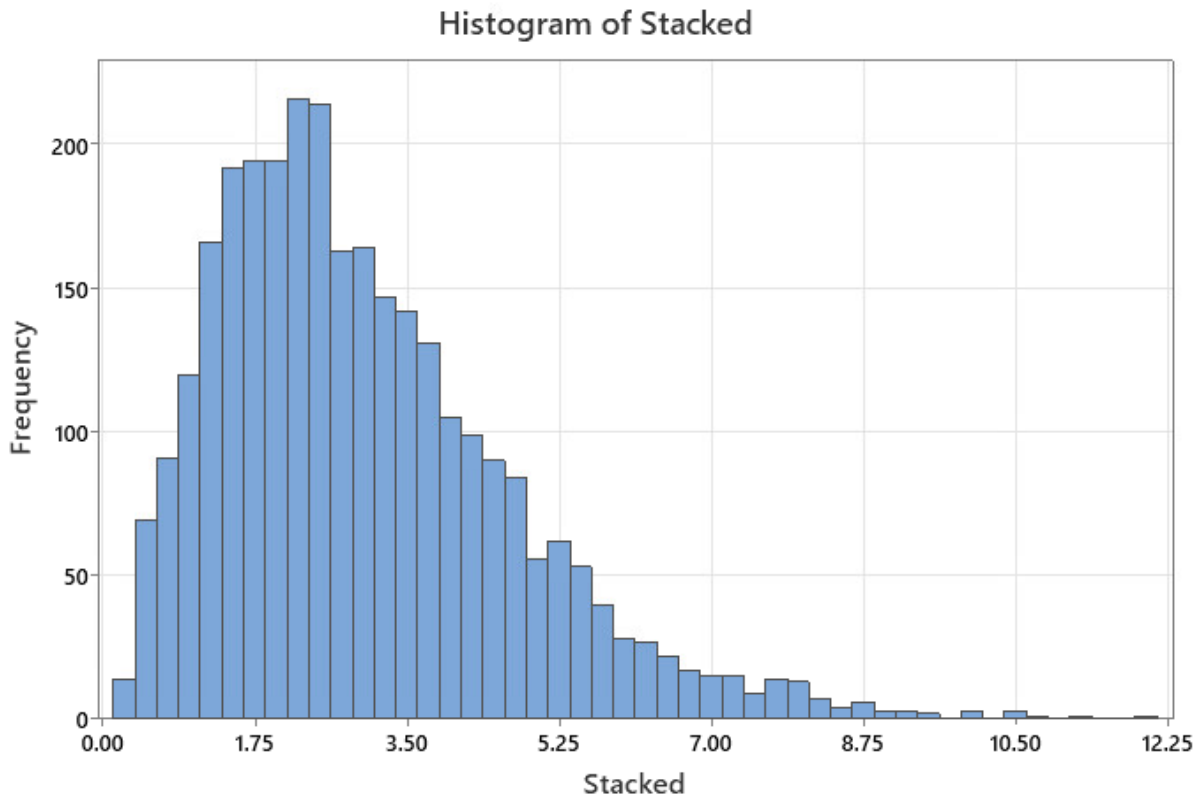



*Figure 13. Exercise 3 : Histogram of Stacked Data for n=30*

This is unchanged from the earlier trials and serves as the benchmark against which the subgroup mean distributions are compared.

For the subgroup means with $n = 30$, the histogram in Figure 14 shows a distribution that is almost perfectly symmetric and bell-shaped.

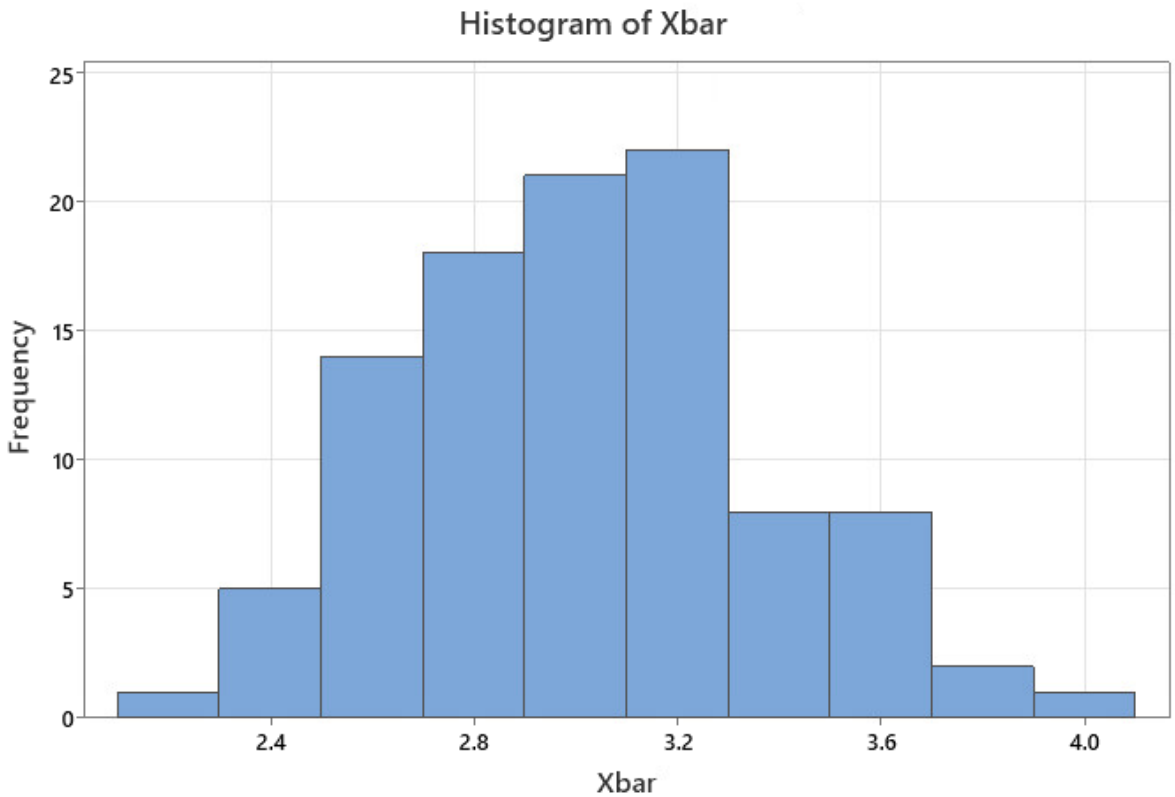



*Figure 14. Exercise 3 : Histogram of Sample Averages for n=30*

The means cluster tightly around 3.0, with far less spread than was observed for $n = 5$and noticeably less than for $n = 10$. This clear reduction in variability reflects the effect of increasing subgroup size: as $n$increases, the standard error of the mean decreases, causing the distribution of averages to narrow and stabilize.

The probability plots of the subgroup means in Figure 15 reinforce this pattern.

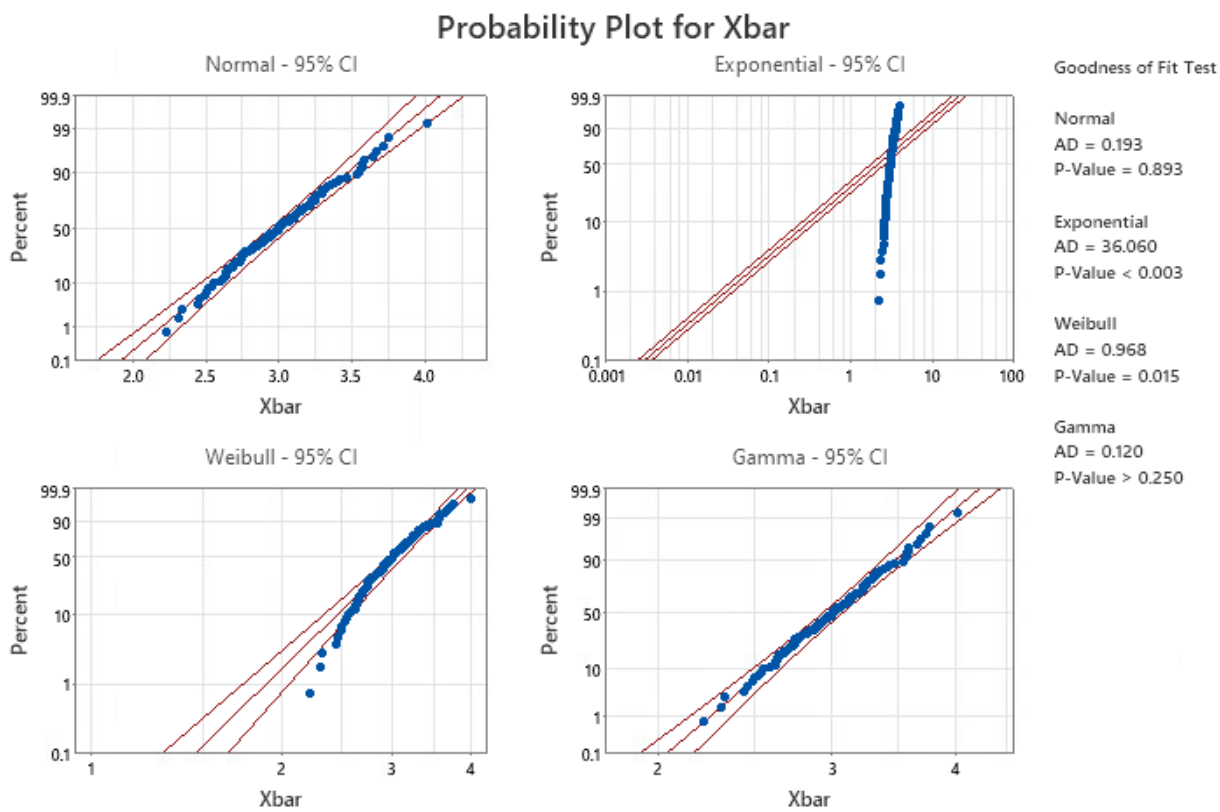



*Figure 15. Exercise 3 : Probability Plot Comparison for n=30*

At $n = 5$, the normal fit was reasonable but not ideal, and at $n = 10$the alignment improved but still left some departure from linearity. By contrast, at $n = 30$the normal probability plot shows points falling almost exactly on the reference line, with a high p-value of 0.893 that strongly supports normality. The gamma distribution continues to fit well, but the improvement of the normal fit with increasing $n$provides the clearest evidence of the Central Limit Theorem.

# 4 Conclusions

The results of the lab confirm that statistical methods and distributional models are effective tools for evaluating process quality and behavior. In the first exercise, descriptive statistics and control charts established that the process was stable, normally distributed, and capable of producing values within specification limits. In the second exercise, theoretical calculations for the mean and variance of a linkage length matched closely with simulated values, and the confidence interval for the sample mean confirmed agreement with theoretical predictions. In the third exercise, the Central Limit Theorem was illustrated

by showing that, while the parent Gamma distribution was skewed, the distribution of the sample mean became progressively more normal as the subgroup size increased, with $n = 30$producing a distribution nearly indistinguishable from the normal model. Overall, the lab demonstrated how statistical analysis can be used both to validate theoretical expectations and to provide evidence of process behavior in practical scenarios.